



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 1

2017 HSC Course Assessment Task 4 (Trial HSC)

Monday August 7, 2017

General instructions

- Working time – 2 hours.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order after this paper, tie into one bundle with the string provided and hand to examination supervisors.
- A NESA Reference Sheet is provided.

SECTION I

- Mark your answers on the answer grid provided (on page 9)

SECTION II

- Commence each new question on a new booklet. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

NESA STUDENT #: # BOOKLETS USED:

Class (please ✓)

☐ 12MAT.1 – Mrs Bhamra

☐ 12MAT.3 – Mr Wall

☐ 12MAT.4 – Mr Sekaran

☐ 12MAT.2 – Mr Lam

☐ 12MAT.5 – Mrs Gan

Marker's use only.

QUESTION	1-10	11	12	13	14	Total	%
MARKS	$\overline{10}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{70}$	

Section I

10 marks

Attempt Question 1 to 10

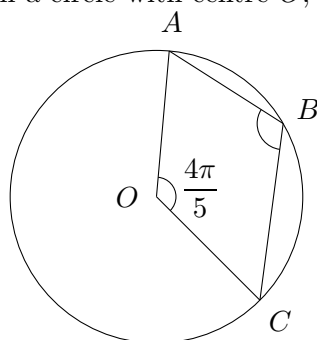
Allow approximately 10 minutes for this section

Mark your answers on the answer grid provided (labelled as page 9).

Questions

Marks

1. The points A , B and C lie on a circle with centre O , as shown in the diagram. 1



The size of $\angle AOC$ is $\frac{4\pi}{5}$ radians.

What is the size of $\angle ABC$ in radians?

- (A) $\frac{3\pi}{10}$ (B) $\frac{\pi}{2}$ (C) $\frac{3\pi}{5}$ (D) $\frac{4\pi}{5}$

2. Which of the following is the range of the function 1

$$y = 2 \sin^{-1} x + \frac{\pi}{2}$$

- (A) $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$ (C) $-\pi \leq y \leq \pi$

- (B) $-\frac{\pi}{2} \leq y \leq \frac{3\pi}{2}$ (D) $-\pi \leq y \leq \frac{3\pi}{2}$

3. What is the y coordinate of the point that divides the interval joining $P(-2, 2)$ and $Q(8, -3)$ internally in the ratio $3 : 2$? 1

- (A) 2 (B) 1 (C) 0 (D) -1

4. Which of the following is the derivative of $\sin^{-1} \frac{2x}{3}$? 1

- (A) $\frac{3}{\sqrt{9-4x^2}}$ (B) $\frac{2}{\sqrt{9-4x^2}}$ (C) $\frac{3}{\sqrt{3-2x^2}}$ (D) $\frac{2}{\sqrt{3-2x^2}}$

5. What is the value of $\lim_{x \rightarrow 0} \frac{\sin x \cos x}{2x}$? 1

- (A) 2 (B) 1 (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

6. Which of the following is a correct intermediate step in evaluating

1

$$\int_0^1 3x(x^2 + 1)^5 dx$$

by using the substitution $u = x^2 + 1$?

- (A) $3 \int_1^2 u^5 du$ (B) $\frac{3}{2} \int_1^2 u^5 du$ (C) $3 \int_0^1 u^5 du$ (D) $\frac{3}{2} \int_0^1 u^5 du$

7. A particle undergoing simple harmonic motion in a straight line has an acceleration of $\ddot{x} = 25 - 5x$, where x is the displacement after t seconds.

Where is the centre of motion?

- (A) $x = 0$ (B) $x = 5$ (C) $x = 10$ (D) $x = 15$

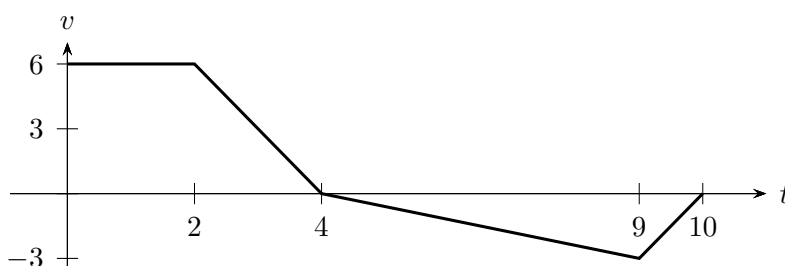
8. The function $f(x) = \sin x - \frac{2x}{3}$ has a real root close to $x = 1.5$. If $x = 1.5$ is the first approximation, what is the next approximation to the root by using Newton's Method of approximation?

1

- (A) 1.495 (B) 1.496 (C) 1.503 (D) 1.504

9. The diagram below shows the velocity-time graph of an object that moves over a ten second time interval.

1



For what percentage of time is the speed of the object decreasing?

- (A) 30% (C) 70%
(B) 60% (D) Cannot determine from the graph.

10. What is the solution of the inequation $3x + 2 < |2x - 1|$?

1

- (A) $x < -\frac{1}{5}$ (C) $-3 < x < \frac{1}{5}$
(B) $x < -3$ (D) $x < -\frac{1}{5}$ or $x > 3$

Examination continues overleaf...

Section II

60 marks

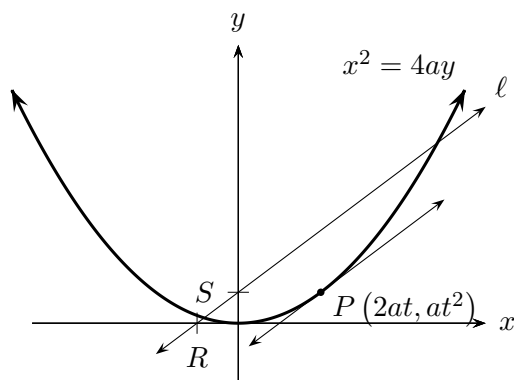
Attempt Questions 11 to 14

Allow approximately 1 hour and 50 minutes for this section.

Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 Marks)	Commence a NEW booklet.	Marks
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- | | | |
|-----|---|----------|
| (a) | Solve for x : $\frac{x}{x^2 - 1} > 0$ | 3 |
| (b) | Find the size of the acute angle between the lines $x - y = 2$ and $2x + y = 1$.
Give your answer to the nearest degree. | 2 |
| (c) | $P(2at, at^2)$ is any point on the parabola $x^2 = 4ay$. The line ℓ is parallel to the tangent at P and passes through the focus S of the parabola. | |



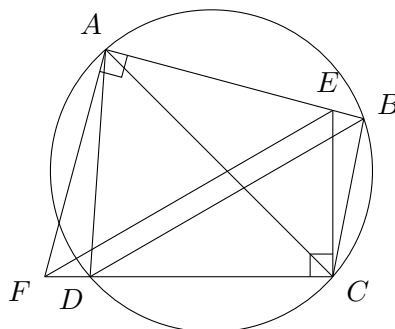
- | | | |
|------|--|----------|
| i. | Find the equation of the line ℓ . | 3 |
| ii. | The line ℓ intersects the x axis at the point R . Find the coordinates of the midpoint M , of the interval RS . | 2 |
| iii. | Find the equation of the locus of M as the point P moves along the parabola where $t \neq 0$. | 1 |

(Note: Consideration of the case where $t = 0$ is not required for purposes of this question.)

Question 11 continues overleaf ...

Question 11 continued from page 7...

- (d) $ABCD$ is a cyclic quadrilateral with $\angle FAE = \angle ECD = 90^\circ$.



Copy or trace the diagram into your writing booklet.

- i. Give a reason why $AECF$ is a cyclic quadrilateral. 1
- ii. Hence prove that EF is parallel to BD . 3

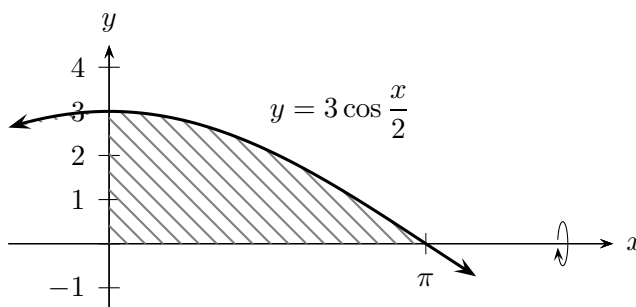
Question 12 (15 Marks)

Commence a NEW booklet.

Marks

- (a)
 - i. Show that $\cos(A - B) = \cos A \cos B (1 + \tan A \tan B)$. 1
 - ii. Suppose that $0 < B < \frac{\pi}{2}$ and $B < A < \pi$. 1
Deduce that if $\tan A \tan B = -1$, then $A - B = \frac{\pi}{2}$.

- (b) The region bounded by the graph $y = 3 \cos \frac{x}{2}$ and the x axis between $x = 0$ and $x = \pi$ is rotated about the x axis to form a solid. 3



Find the exact volume of the solid.

- (c) Prove by mathematical induction that $5^n + 12n - 1$ is divisible by 16 for all positive integers $n \geq 1$. 3
- (d) Find $\int \frac{1}{x^2 + 2x + 2} dx$ 2
- (e) Consider the function $f(x) = \frac{x}{x+4}$.
- i. Explain why $f(x)$ has an inverse function $f^{-1}(x)$. 1
 - ii. Find an expression for the inverse function $f^{-1}(x)$. 2
 - iii. Find the point(s) of intersection of $y = f(x)$ and $y = f^{-1}(x)$. 2

Question 13 (15 Marks)

Commence a NEW booklet.

Marks

- (a) Newton's Law of Cooling states that when an object at temperature $T^\circ\text{C}$ is placed in an environment $T_0^\circ\text{C}$, the rate of the temperature loss is given by the equation

$$\frac{dT}{dt} = -k(T - T_0)$$

where t is the time in minutes, and k is a positive constant.

An object whose initial temperature is 90°C is placed in a room in which the internal temperature is maintained at 20°C . After 10 minutes, the temperature of the object is 70°C .

- i. Show that $T = T_0 + Ae^{-kt}$ satisfies the above equation. **1**
 - ii. Show that $k = \frac{1}{10} \log_e \frac{7}{5}$. **1**
 - iii. How long will it take for the object's temperature to reduce to 63°C ? **2**
- (b) A particle moves in a straight line such that its acceleration $\ddot{x} \text{ ms}^{-2}$ is given by

$$\ddot{x} = x + \frac{3}{2}$$

Initially, the particle was 5 metres to the right of O and moving towards O with a speed of 6 ms^{-1} .

- i. Initially was the particle speeding up or slowing down? Justify your answer. **1**
 - ii. Show that $v^2 = x^2 + 3x - 4$. **2**
 - iii. Where does the particle first change direction? **1**
- (c) A particle moves in a straight line and its position at time t is given by

$$x = 5 + \sqrt{3} \sin 3t - \cos 3t$$

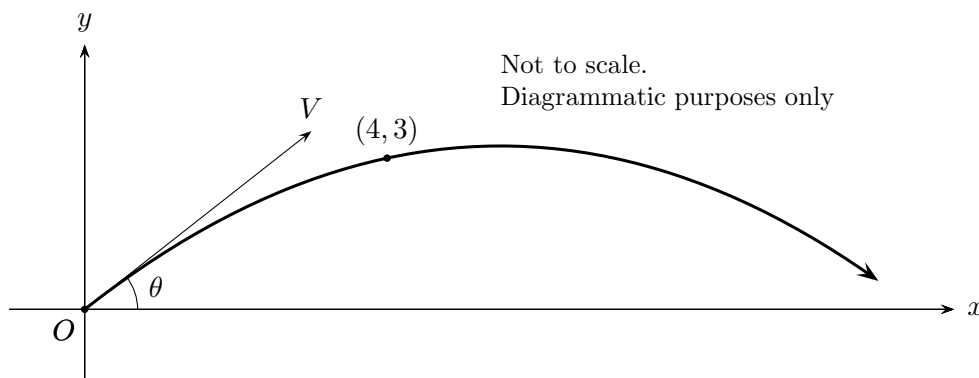
- i. Express $\sqrt{3} \sin 3t - \cos 3t$ in the form $R \sin(3t - \alpha)$, where α is in radians. **2**
- ii. Prove that the particle is undergoing simple harmonic motion. **1**
- iii. Find the amplitude and centre of motion. **2**
- iv. Find the first time when the particle is at its minimum displacement. **2**

Question 14 (15 Marks)

Commence a NEW booklet.

Marks

- (a) A particle is projected from a point O with speed of V metres per second at an angle of θ to the horizontal. Air resistance is negligible, and the acceleration due to gravity is $g \text{ ms}^{-2}$. The particle also passes through the point $(4, 3)$.



The displacement-time equations for the projectile are

$$x = Vt \cos \theta \quad (\text{Do NOT prove these})$$

$$y = -\frac{1}{2}gt^2 + Vt \sin \theta$$

- i. Show that the Cartesian equation of the trajectory is given by

2

$$y = x \tan \theta - \frac{gx^2}{2V^2} (1 + \tan^2 \theta)$$

- ii. Find the initial angle of projection θ to the nearest minute, if $V^2 = 8g$.

2

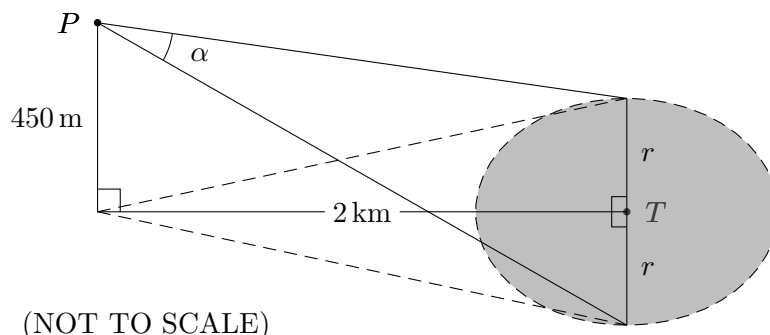
- iii. Find the range of the projectile.

1

Question 14 continues overleaf ...

Question 14 continued from page 7...

- (b) An oil tanker at T is leaking oil which forms a circular oil slick. An observer is measuring the oil slick from a position P , 450 metres above sea level and 2 kilometres horizontally from the centre of the oil slick.



(NOT TO SCALE)

- i. At a certain time the observer measures the angle, α , subtended by the diameter of the oil slick, to be 0.2 radians. What is the radius, r , at this time? **2**
 - ii. At this time, $\frac{d\alpha}{dt} = 0.04$ radians per hour. Find the rate at which the area of the oil slick is growing. **3**
- (c) It is given that $P(x) = (x - a)^3 + (x - b)^2$. The remainder when $P(x)$ is divided by $(x - b)$ is -8 .
- i. Show that when $P(x)$ is divided by $(x - a)$, the remainder is 4. **1**
 - ii. Prove that $x = \frac{a + b}{2}$ is a zero of $P(x)$. **1**
 - iii. Prove that $P(x)$ has no stationary points. **3**

End of paper.

Sample Band E4 Responses

Section I

1. (C) 2. (B) 3. (D) 4. (B) 5. (C)
6. (B) 7. (B) 8. (B) 9. (A) 10. (A)

Section II

Question 11 (Lam)

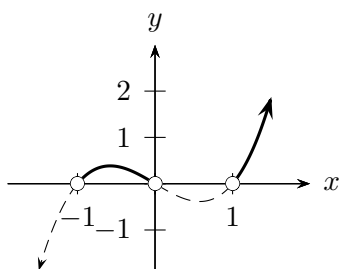
(a) (3 marks)

- ✓ [1] for correctly multiplying by the square of the denominator.
- ✓ [1] diagram or otherwise in assisting with the final inequalities.
- ✓ [1] for correct inequalities.

$$\frac{x}{x^2 - 1} > \frac{0}{x(x^2 - 1)^2}$$

$$x(x^2 - 1) > 0$$

$$x(x - 1)(x + 1) > 0$$

Hence $-1 < x < 0$ or $x > 1$.

(b) (2 marks)

- ✓ [1] for correct $\tan(\alpha - \beta)$ formula.
- ✓ [1] for final answer.

$$\begin{aligned} x - y = 2 &\Rightarrow m_1 = 1 \\ 2x + y = 1 &\Rightarrow m_2 = -2 \\ \tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\ &= \frac{1 - (-2)}{1 + (1)(-2)} \\ &= |-3| = 3 \\ \therefore \theta &\approx 72^\circ \end{aligned}$$

(c) i. (3 marks)

- ✓ [1] for correctly differentiating $y = \frac{x^2}{4a}$.
- ✓ [1] for the correct gradient.
- ✓ [1] for correct equation.

$$\begin{aligned} x^2 &= 4ay \\ y &= \frac{x^2}{4a} \\ \frac{dy}{dx} &= \frac{2x}{4a} \Big|_{x=2at} = \frac{4at}{4a} = t \\ \therefore y &= tx + b \end{aligned}$$

As the line ℓ passes through the focus $(0, a)$,

$$\therefore y = tx + a$$

ii. (2 marks)

- ✓ [1] for each correct coordinate of M .

$$y = tx + a$$

At R , $y = 0$:

$$\begin{aligned} tx + a &= 0 \\ \therefore x_R &= -\frac{a}{t} \\ \therefore R &\left(-\frac{a}{t}, 0\right) \end{aligned}$$

The midpoint of RS :

$$\begin{aligned} M &= \left(\frac{-\frac{a}{t} + 0}{2}, \frac{0 + a}{2} \right) \\ &= \left(-\frac{a}{2t}, \frac{a}{2} \right) \end{aligned}$$

iii. (1 mark)

$$y = \frac{a}{2}$$

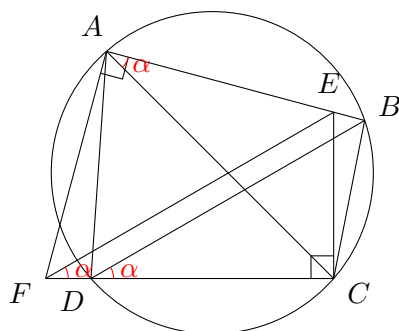
(as y is independent of t)

(d) i. (1 mark)

- Opposite angles $\angle FAC$ and $\angle ECF$ are supplementary (as they are both 90°).

ii. (3 marks)

- ✓ [1] for finding angles in the same segment (twice)
- ✓ [1] for final result, based on correct argument.



- $\angle BDC = \angle BAC$ (angles in the same segment, circle $ABCD$)
- $\angle EAC = \angle EFC$ (angles in the same segment, circle $AECF$)

Hence $\angle BDC = \angle EFC$, which are corresponding angles that are equal.
Hence $EF \parallel BD$.

Question 12 (Bhamra)

(a) i. (1 mark)

$$\begin{aligned} & \cos(A - B) \\ &= \cos A \cos B + \sin A \sin B \\ &= \cos A \cos B \left(1 + \frac{\sin A \sin B}{\cos A \cos B} \right) \\ &= \cos A \cos B (1 + \tan A \tan B) \end{aligned}$$

ii. (1 mark)

$$\begin{aligned} \cos(A - B) &= \cos A \cos B (1 + \tan A \tan B) \\ &= \cos A \cos B (1 + (-1)) \\ &= 0 \\ \therefore A - B &= \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2} \dots = (2k + 1) \frac{\pi}{2} \end{aligned}$$

Since $A < \pi$ and $B > 0$,

$$\begin{aligned} \therefore A - B &< \pi \\ \therefore A - B &= \frac{\pi}{2} \end{aligned}$$

(b) (3 marks)

- ✓ [1] for correct volume integral.
- ✓ [1] for correct primitive.
- ✓ [1] for correct final answer.

$$\begin{aligned} V &= \pi \int_0^\pi 9 \cos^2 \frac{x}{2} dx \\ &= 9\pi \int_0^\pi \left(\frac{1}{2} + \frac{1}{2} \cos x \right) dx \\ &= \frac{9\pi}{2} [x + \sin x]_0^\pi \\ &= \frac{9\pi}{2} ((\pi + \sin \pi) - (0 + 0)) \\ &= \frac{9\pi^2}{2} \end{aligned}$$

(c) (3 marks)

- ✓ [1] for correctly proving the base case.
- ✓ [1] for using the base case in the inductive hypothesis.
- ✓ [1] for final proof.

Let $P(n)$ be the proposition $5^n + 12n - 1$ is divisible by 16, i.e.

$$5^n + 12n - 1 = 16M$$

- $P(1)$:

$$5^1 + 12 - 1 = 17 - 1 = 16$$

Hence $P(1)$ is true.

- Inductive step: assume $P(k)$ is true, $k \in \mathbb{Z}^+$, i.e.

$$5^k + 12k - 1 = 16P$$

is true.

- Examine $P(k + 1)$:

$$\begin{aligned} & 5^{k+1} + 12(k + 1) - 1 \\ &= 5 \times 5^k + 12k + 11 \\ &= 5(16P - 12k + 1) + 12k + 11 \\ &= 16 \times 5P - 60k + 5 + 12k + 11 \\ &= 16 \times 5P - 48k + 16 \\ &= 16(5P - 3k + 1) \\ &= 16Q \end{aligned}$$

Hence $P(k + 1)$ is also true, and $P(n)$ is true by induction.

(d) (2 marks)

$$\begin{aligned}\int \frac{1}{x^2 + 2x + 2} dx &= \int \frac{1}{(x+1)^2 + 1} dx \\ &= \tan^{-1}(x+1) + C\end{aligned}$$

(e) i. (1 mark)

$$\begin{aligned}f(x) &= \frac{x}{x+4} = \frac{x+4-4}{x+4} \\ &= 1 - \frac{4}{x+4}\end{aligned}$$

i.e. a regular hyperbola which is one-to-one (monotonic increasing). Hence $f(x)$ has an inverse function.

ii. (2 marks)

- ✓ [1] for interchanging variables.
✓ [1] for final answer.

$$f : y = 1 - \frac{4}{x+4}$$

Interchanging variables,

$$\begin{aligned}x &= 1 - \frac{4}{y+4} \\ x-1 &= -\frac{4}{y+4} \\ y+4 &= -\frac{4}{x-1} \\ y &= -4 - \frac{4}{x-1}\end{aligned}$$

iii. (2 marks)

- ✓ [1] for realising f and f^{-1} intersect along $y = x$.
✓ [1] for both points of intersection. $f(x)$ and $f^{-1}(x)$ intersect along $y = x$:

$$\begin{aligned}x &= \frac{x}{x+4} \\ x - \frac{x}{x+4} &= 0 \\ x \left(1 - \frac{1}{x+4} \right) &= 0 \\ x = 0 \quad \text{or} \quad \frac{1}{x+4} &= 1 \\ \therefore x = 0 \text{ or } x = -3\end{aligned}$$

Points of intersection are (0,0) and (-3,-3).

Question 13 (Sekaran)

(a) i. (1 mark)

$$\begin{aligned}T &= T_0 + Ae^{-kt} \\ \therefore T - T_0 &= Ae^{-kt}\end{aligned}$$

Differentiating,

$$\frac{dT}{dt} = -kAe^{-kt} = -k(T - T_0)$$

Hence $T = T_0 + Ae^{-kt}$ satisfies the equation $\frac{dT}{dt} = -k(T - T_0)$.

ii. (1 mark)

$$\begin{aligned}t = 0, T &= 90 \\ \therefore 90 &= 20 + Ae^0 \\ A &= 70 \\ T &= 20 + 70e^{-kt}\end{aligned}$$

When $t = 10$, $T = 70$:

$$\begin{aligned}70 &= 20 + 70e^{-10k} \\ e^{-10k} &= \frac{5}{7} \\ -10k &= \ln \frac{5}{7} \\ k &= -\frac{1}{10} \ln \frac{5}{7} = \frac{1}{10} \ln \frac{7}{5}\end{aligned}$$

iii. (2 marks)

- ✓ [1] for obtaining expression for e^{-kt} .
✓ [1] for final answer.

$$T = 20 + 70e^{-kt}$$

Finding t when $T = 63$:

$$\begin{aligned}63 &= 20 + 70e^{-kt} \\ e^{-kt} &= \frac{43}{70} \\ -kt &= \ln \frac{43}{70} \\ t &= \frac{1}{k} \ln \frac{70}{43} \\ &= \frac{10 \ln \frac{70}{43}}{\ln \frac{7}{5}} \\ &\approx 14.482 \dots\end{aligned}$$

(15 minutes)

(b) i. (1 mark)

$$\ddot{x} = x + \frac{3}{2}$$

When $t = 0$, $x = 5$:

$$\ddot{x} = 5 + \frac{3}{2} = 6.5 \text{ ms}^{-2} > 0$$

$$\dot{x} = -6 \text{ ms}^{-1} < 0$$

Hence the particle is slowing down as $\ddot{x}$ and $\dot{x}$ have opposite signs.

ii. (2 marks)

✓ [1] for obtaining the primitive which is equal to $\frac{1}{2}v^2$

✓ [1] for final answer.

$$\ddot{x} = \frac{d}{dx} \left(\frac{1}{2}v^2 \right) = x + \frac{3}{2}$$

$$\therefore \frac{1}{2}v^2 = \int \left(x + \frac{3}{2} \right) dx$$

$$= \frac{1}{2}x^2 + \frac{3}{2}x + C_1$$

$$v^2 = x^2 + 3x + C_2$$

When $t = 0$, $x = 5$, $v = -6$:

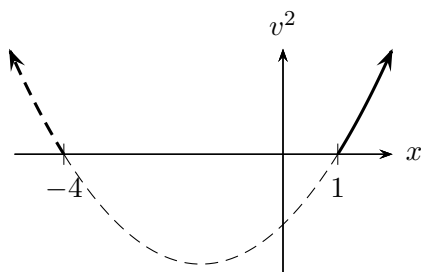
$$(-6)^2 = 5^2 + 3(5) + C_2$$

$$C_2 = -4$$

$$\therefore v^2 = x^2 + 3x - 4$$

iii. (1 mark)

$$v^2 = x^2 + 3x - 4 = (x - 1)(x + 4)$$



Particle commences at $x = 5$ and is therefore relegated to the right branch of the parabolic $v^2 - x$ graph. At $x = 1$, $v = 0$ and $\ddot{x} = 2.5 > 0$. Hence the particle first changes direction when it is 1 metre to the right of O .

(c) i. (2 marks)

$$\sqrt{3} \sin 3t - \cos 3t$$

$$= R \sin(3t - \alpha)$$

$$= R \sin 3t \cos \alpha - R \cos 3t \sin \alpha$$

Comparing coefficients,

$$\begin{cases} R \cos \alpha = \sqrt{3} & (1) \\ R \sin \alpha = 1 & (2) \end{cases}$$

(2) $\div$ (1):

$$\therefore \tan \alpha = \frac{1}{\sqrt{3}}$$

$$\alpha = \frac{\pi}{6}$$

$$R \sin \frac{\pi}{6} = 1$$

$$\therefore R = 2$$

$$\therefore \sqrt{3} \sin 3t - \cos 3t = 2 \sin \left(3t - \frac{\pi}{6} \right)$$

ii. (1 mark)

$$x = 5 + \sqrt{3} \sin 3t - \cos 3t$$

$$= 5 + 2 \sin \left(3t - \frac{\pi}{6} \right)$$

$$\dot{x} = 2 \times 3 \cos \left(3t - \frac{\pi}{6} \right)$$

$$\ddot{x} = -2 \times 3^2 \sin \left(3t - \frac{\pi}{6} \right)$$

$$= -9 \left(2 \sin \left(3t - \frac{\pi}{6} \right) \right)$$

$$= -9(x - 5)$$

As acceleration is proportional to but directed against the displacement from the centre of motion $x = 5$, the particle is undergoing SHM.

iii. (2 marks)

$$a = 2 \quad x_C = 5$$

iv. (2 marks)

✓ [1] for recognition when the minimum displacement occurs.

✓ [1] for final answer.

Minimum displacement occurs when

$$\sin\left(3t - \frac{\pi}{6}\right) = -1:$$

$$\sin\left(3t - \frac{\pi}{6}\right) = -1$$

$$3t - \frac{\pi}{6} = \frac{3\pi}{2}, \dots$$

$$3t = \frac{10\pi}{6} = \frac{5\pi}{3}$$

$$\therefore t = \frac{5\pi}{9}$$

Question 14 (Gan)

(a) i. (2 marks)

$$\begin{cases} x = Vt \cos \theta & (1) \\ y = -\frac{1}{2}gt^2 + Vt \sin \theta & (2) \end{cases}$$

From (1):

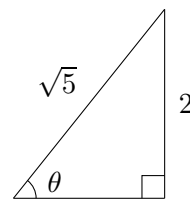
$$t = \frac{x}{V \cos \theta}$$

Substitute into (2):

$$\begin{aligned} y &= -\frac{1}{2}g \left(\frac{x}{V \cos \theta}\right)^2 + V \left(\frac{x}{V \cos \theta}\right) \sin \theta \\ &= -\frac{gx^2}{2V^2} \sec^2 \theta + x \tan \theta \\ &= -\frac{gx^2}{2V^2} (1 + \tan^2 \theta) + x \tan \theta \end{aligned}$$

ii. (2 marks) Using $V^2 = 8g$ and substituting $x = 4$, $y = 3$:

$$\begin{aligned} 3 &= -\frac{g \times 16}{2 \times 8g} (1 + \tan^2 \theta) + 4 \tan \theta \\ 3 &= -1 - \tan^2 \theta + 4 \tan \theta \\ \tan^2 \theta - 4 \tan \theta + 4 &= 0 \\ (\tan \theta - 2)^2 &= 0 \\ \tan \theta &= 2 \\ \theta &\approx 63^\circ 26' \end{aligned}$$

iii. (1 mark) At the range R , $y = 0$. Also, draw picture depicting $\tan \theta = 2$:

$$\text{i.e. } \sec \theta = \sqrt{5} \quad 1$$

$$0 = -\frac{gx^2}{16g} \sec^2 \theta + x \tan \theta$$

$$0 = -\frac{x^2}{16} \times 5 + 2x$$

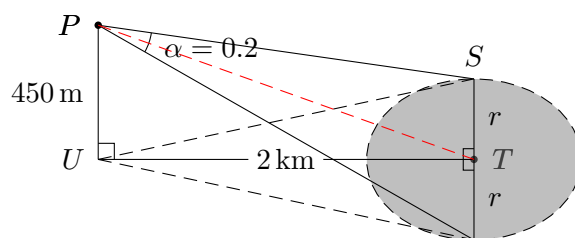
$$x \left(2 - \frac{5}{16}x\right) = 0$$

As $x \neq 0$,

$$\frac{5}{16}x = 2$$

$$x = \frac{32}{5} = 6.4 \text{ m}$$

(b) i. (2 marks)

• In $\triangle PUT$,

$$\begin{aligned} PT^2 &= 450^2 + 2000^2 \\ \therefore PT &= 2050 \end{aligned}$$

• In $\triangle PTS$,

$$\begin{aligned} \tan 0.1 &= \frac{r}{PT} = \frac{r}{2050} \\ \therefore r &= 2050 \tan 0.1 \approx 205.69 \end{aligned}$$

ii. (3 marks)

✓ [1] for obtaining $\frac{dA}{d\alpha}$.✓ [1] for substituting in values for $\frac{dA}{d\alpha} \times \frac{d\alpha}{dt}$.

✓ [1] for final answer.

As $A = \pi r^2$,

$$A = \pi \times 2050^2 \tan^2 \frac{\alpha}{2}$$

$$\begin{aligned} \frac{dA}{d\alpha} &= 2050^2 \times \pi \times \sec^2 \frac{\alpha}{2} \times \tan \frac{\alpha}{2} \\ &= 2050^2 \times \pi \times \sec^2 0.1 \times \tan 0.1 \end{aligned}$$

Applying the chain rule,

$$\begin{aligned}\frac{dA}{dt} &= \frac{dA}{d\alpha} \times \frac{d\alpha}{dt} \\ &= 2050^2 \times \pi \times \sec^2 0.1 \times \tan 0.1 \times 0.04 \\ &= \frac{2050^2 \pi \tan 0.1}{\cos^2 0.1} \times 0.04 \\ &\approx 53\,520.33 \text{ square metres per hour}\end{aligned}$$

(c) i. (1 mark)

$$\begin{aligned}P(x) &= (x-a)^3 + (x-b)^2 \\ P(b) &= -8 \\ \therefore (b-a)^3 + (b-b)^2 &= -8 \\ (b-a)^3 &= -8 \\ b-a &= -2 \quad (\dagger)\end{aligned}$$

Applying the remainder theorem and evaluating $P(a)$:

$$\begin{aligned}P(a) &= (a-a)^3 + (a-b)^2 \\ &= (a-b)^2 \\ &= 4\end{aligned}$$

Hence the remainder when divided by $(x-a)$, is 4.

ii. (1 mark)

$$\begin{aligned}P\left(\frac{a+b}{2}\right) &= \left(\frac{a+b}{2} - a\right)^3 + \left(\frac{a+b}{2} - b\right)^2 \\ &= \left(\frac{a+b-2a}{2}\right)^3 + \left(\frac{a+b-2b}{2}\right)^2 \\ &= \left(\frac{b-a}{2}\right)^3 + \left(\frac{a-b}{2}\right)^2 \\ &= \left(-\frac{2}{2}\right)^3 + \left(\frac{2}{2}\right)^2 \\ &= -1 + 1 = 0\end{aligned}$$

Hence $x = \frac{a+b}{2}$ is a zero of $P(x)$.

iii. (3 marks)

- ✓ [1] for obtaining $3(x-a)^2 + 2(x-b) = 0$.
- ✓ [1] for correct substitution into Δ .
- ✓ [1] for correct proof.

$$P(x) = (x-a)^3 + (x-b)^2$$

Differentiating,

$$P'(x) = 3(x-a)^2 + 2(x-b)$$

Stationary points occur when $P'(x) = 0$:

$$\begin{aligned}3(x-a)^2 + 2(x-b) &= 0 \\ 3x^2 - 6ax + 3a^2 + 2x - 2b &= 0 \\ 3x^2 + (2-6a)x + (3a^2-2b) &= 0\end{aligned}$$

Checking the discriminant of this quadratic:

$$\begin{aligned}\Delta &= (2-6a)^2 - 4(3)(3a^2-2b) \\ &= 4 - 24a + 36a^2 - 36a^2 + 24b \\ &= 4 - 24(a-b) \\ &= 4 - 24(2) \\ &< 0\end{aligned}$$

$P'(x) = 0$ has no real roots. Hence $P(x)$ has no stationary points.